

Neovaginal malignant melanoma following surgery and radiation for vulvar squamous cell carcinoma.

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The causal link between therapeutic irradiation and malignant melanoma has not been firmly established. The authors report a novel case of a malignant melanoma arising from a therapeutically irradiated neovagina reconstructed from skin flaps following radical vulvectomy with inguinal lymph node dissection 16 years earlier. The diagnosis was established by light microscopy and immunohistochemical staining. Reports of melanoma arising from sites of previous radiation therapy are reviewed. Our observations suggest that therapeutic radiation may be a factor in the development of malignant melanoma, particularly in non-sun-exposed areas, including the genital tract.

OrienX010, an oncolytic virus, in patients with unresectable stage IIIC-IV melanoma: a phase Ib study.

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Melanoma in people of Asian descent presents primarily in non-sun-exposed areas, such as acral and mucosal melanoma. Compared with the predominant sun-exposed area melanomas in Caucasians, acral and mucosal melanomas do not respond as well to immunotherapy and are associated with a worse prognosis. Hence, there is an urgent need for improved treatment for melanoma in Asians. This phase Ib trial evaluated the safety and efficacy of the modified herpes simplex virus-1 oncolytic virus OrienX010 in Chinese patients with unresectable stage IIIC-IV melanoma. Patients were treated in two different cohorts. In cohort 08 (n=12), patients received up to 5 mL of 8×10^7 pfu/mL OrienX010 intratumoral injections every 2 weeks until disease progression and responses were evaluated every 6 weeks. In cohort 09 (n=14), patients received up to 10 mL of 8×10^7 pfu/mL OrienX010 intratumoral injections and responses were evaluated every 8 weeks. Between June 2014 and May 2017, 26 patients were enrolled, including 18 (69.2%) patients with acral melanoma. Fever and injection site reaction were the most frequent adverse events. Only one patient experienced a grade ≥ 3 adverse event and no dose-limiting toxicities were observed. The objective response rate was 19.2% and the disease control rate was 53.8%. The median duration of response was 6.0 months. Antitumor effects were observed in 54.6% of injected lesions and 48.8% of non-injected lesions, including one (16.7%) of six evaluable distant lung metastases. The median progression-free survival was 2.9 months and overall survival was 19.2 months. Compared with patients treated in cohort 08, patients treated in cohort 09 had an improved objective response rate (28.6% vs 8.3%) and a median progression-free survival of 3.0 months vs 2.8 months. OrienX010 oncolytic virotherapy has a tolerable safety profile with antitumor effects in both injected and non-injected metastases and warrants further evaluation in patients with melanoma. Based on these results, the higher cohort 09 dose (up to 10 mL of 8×10^7 pfu/mL every 2 weeks) was selected as the recommended phase II dose for ongoing trials. CTR20140631 (cohort 08), CTR20150881 (cohort 09).

Human herpes virus DNA is rarely detected in non-UV light-associated primary malignant melanomas of mucous membranes.

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UV-radiation is the most important causative factor for malignant melanomas of the skin. However, this is not the case for melanomas on sun-sheltered body surfaces. The aim of this study was to investigate if human herpes virus DNA could be found in malignant melanomas in sun-sheltered body areas and if these viruses play a role in the development of extracutaneous melanomas. Forty-one extracutaneous melanomas were dissected and used for further analysis. Quantitative PCR methods were used for detection of the eight human herpes viruses in melanoma samples. Human herpes virus DNA was absent in 37/41 melanomas, however, cytomegalovirus DNA was detected in two samples, and one sample each exhibited presence of Epstein-Barr virus and Human Herpes virus-6 DNA respectively. Human herpes virus DNA is rarely detected in primary malignant melanomas in non-sun exposed body surfaces and is not a major factor for the development of extracutaneous melanomas.

Distinct genomic features in a retrospective cohort of mucosal, acral, and vulvovaginal melanomas.

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Compared with sun-exposed melanomas, less is known regarding the pathogenesis of sun-protected melanomas. Sun-protected melanomas share many epidemiologic factors, but their genetic heterogeneity is not well studied. We investigated the genomic profile of acral, mucosal, and vulvovaginal melanomas. We hypothesize that mucosal melanomas, recognized for their uniquely aggressive clinical behavior, have distinct genomic features. We performed whole transcriptome messenger RNA and DNA (1711 genes) sequencing, messenger RNA expression profiling, tumor mutational burden, ultraviolet signature, and copy number variants analysis on 29 volar/digital acral, 7 mucosal, and 6 vulvovaginal melanomas. There was significant genetic heterogeneity, particularly in acral melanomas, with 36% having BRAF alterations, whereas other melanomas had none ($P = .0159$). Nonzero ultraviolet signatures were more frequent in acral melanomas, suggesting greater ultraviolet involvement. Mucosal melanomas formed a distinct group with increased expression of cell cycle and proliferation genes. Various targetable aberrations were identified, such as AURKA and ERBB2, in mucosal and acral melanomas, respectively. The sample size was a small. There is significant genetic heterogeneity among sun-protected melanomas. Mucosal melanomas have upregulation in cell cycle and proliferation genes, which may explain their aggressive behavior. Ultraviolet radiation plays some role in a subset of acral but not other melanomas.

Spatial density of primary malignant melanoma in sun-shielded body sites: A potential guide to melanoma genesis.

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UV radiation is a major factor in melanoma genesis, but non-UV linked factors are also operational, since primary malignant melanomas can emerge in body sites that never see the sun. The scarcity of melanomas in sun-shielded body sites reflects only the absolute number of melanomas, not the number of tumours per square unit of the surface in which they emerge. Studies on melanoma density conducted by us and others are here briefly reviewed. The access to reliable numbers along with measurable anatomical areas directed our choice of melanomas at the sun-shielded locations described here. Melanomas at the body surface. Calculations of surface areas bearing melanomas relative to the total

body surface included sites on the vulva, subungual tissues, volar and palmar skin, and, for comparison melanomas of the face during the same period of time. The density of vulvar melanomas was identical to that in chronically sun-exposed facial skin. Subungual melanomas were almost nine times denser than expected whereas melanomas of palms and soles showed a lower density than expected. Melanomas beneath the body surface. The densities of melanomas in the vagina, anal canal and uvea, were calculated separately and compared to the average density of cutaneous melanomas (CMMs) during the same period of time. Melanomas of the anal canal displayed a density almost twice the average for CMMs, whereas the vaginal melanomas were similar in density to CMMs. In contrast, the density of the uveal melanomas was calculated as 50 and 41 times (men and women, respectively) the average density of CMMs. The high density of some melanomas in sun-shielded body areas indicates the presence of factors underlying the origins of these tumours that seem to be equivalent in strength to UV radiation and also implies that specific anatomical sites favour the emergence and proliferation of melanomas, independent of UV radiation.

North-South gradients of melanomas and non-melanomas: A role of vitamin D?

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Incidence rates of skin cancer increase with decreasing latitude in Norway, as in many other countries with white populations. The latitudinal trends of the incidence rates of skin cancer were studied and compared with data for vitamin D-induced by UV and for vitamin D intake. The north-south gradient for CMM incidence rates on sun exposed skin is much smaller than those for BCC and SCC, and that for BCC is smaller than that for SCC. This indicates that SCC and BCC are mainly due to solar UVB, while UVA may play a significant role for CMM and a smaller role for BCC, since the north-south gradient of annual UVB fluences is larger than that of UVA fluences. However, there is an inverse latitudinal gradient of skin cancer in central Europe. This is probably due to a gradient of skin color, since white skin is an important determinant of increased risk of skin cancer. The role of vitamin D for skin cancer risk is difficult to evaluate, since serum levels of 25-hydroxyvitamin D, as well as vitamin D intakes, are widely different from country to country. Still, epidemiological evidence indicates a role: for melanomas arising on non-sun exposed body localizations (uveal melanomas, melanomas arising in the vulva and perianal/anorectal regions) there appears to be no latitudinal gradient, or, a negative gradient, i.e., increasing rates with decreasing latitude as would be expected if UV-generated vitamin D plays a protective role. Both skin cancer risk and vitamin D photosynthesis decrease with increasing skin darkness.

DNA from BK virus and JC virus and from KI, WU, and MC polyomaviruses as well as from simian virus 40 is not detected in non-UV-light-associated primary malignant melanomas of mucous membranes.

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The single most important causative factor for malignant melanomas of the skin is UV radiation. However, this is not true for melanomas on body surfaces sheltered from the sun; thus, it is important to seek new causative factors of melanoma genesis. Human papillomaviruses and gammaherpesviruses are associated with human skin cancer; for example, human papillomavirus types 5 and 8 are associated

with epidermodysplasia verruciformis, and human herpesvirus 8 is associated with Kaposi's sarcoma. Recently, a newly described human polyomavirus, Merkel cell polyomavirus (MCPyV), has been associated with Merkel cell carcinoma, an unusual form of neurotropic skin cancer. Moreover, melanocytes are of neuroepithelial origin. This background impelled us to investigate if human polyomavirus DNA could play a role in the development of extracutaneous melanomas. Sixty-four extracutaneous melanomas were initially collected and dissected. Of these, 38 could be successfully used for further testing for the presence of the five human polyomaviruses known so far-BK virus (BKV), JC virus (JCV), KI polyomavirus (KIPyV), WU polyomavirus (WUPyV), and MCPyV-and of simian virus 40 (SV40). No polyomavirus DNA could be detected in any of the samples tested by use of a nested PCR detecting BKV, JCV, and SV40; a newly designed PCR detecting KIPyV and WUPyV; or a newly designed PCR for MCPyV. We conclude that since no human polyomavirus DNA was detected in primary malignant melanomas on non-sun-exposed body surfaces, these polyomaviruses presumably are not major factors for the development of extracutaneous melanomas.

Primary malignant melanoma of the vulva--an aggressive tumor for modeling the genesis of non-UV light-associated melanomas.

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Malignant melanomas appear in such sun-shielded areas as the vulva, challenging conventional knowledge that they are associated with UV radiation. Based on 1442 patients with vulvar melanomas the tumors' epidemiology, clinical manifestations, histopathology, molecular genetics, treatment strategies, and prognosis were surveyed. Despite their sun-shielded location and rare incidence, vulvar melanomas were, on average, more dense than melanomas on the body surface and nearly the density of melanomas in chronically sun-exposed skin of the head and neck. Vulvar melanomas differed markedly from cutaneous melanomas, as evidenced by histopathological lesions and molecular genetics. Most melanomas were located on the glabrous skin as opposed to the hairy skin within the vulva and differed significantly in biological properties. The prognosis for the patients was poor, and in the 11 largest studies of surgical strategies, none offered a significant survival advantage. Tumor thickness and ulceration were usually significant predictors of (poor) prognosis in multivariate analyses along with macroscopic amelanosis, angioinvasion or DNA non-diploidy in some reports. Clear-cut biological differences between vulvar and cutaneous melanomas and between melanomas within different vulvar sites provide new paths for extensive research on melanomagenesis and for potential therapies. Additionally, studies of vulvar and other extracutaneous melanomas should characterize subgroups of cutaneous melanomas and identify their cause(s), which are apparently not linked to UV radiation.

Clinicopathological features, vitamin D serological levels and prognosis in cutaneous melanoma of shield-sites: an update.

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Intermittent sun exposure and sunburns are strongly related to the development of melanoma (MM); however, MM can also arise in non-sun exposed areas, where other biological pathways may cause the disease, with different outcomes. At the same time, evidences of serum levels of vitamin D in melanoma patients according to sun-exposed or not-sun-exposed areas are still lacking, especially if compared with

the percentage of BRAF mutation. We performed a retrospective analysis with patients registered in our electronic database and an observational study in patients with a recent diagnosis of MM. Performing Kaplan-Meier product and log-rank test, median disease-free survival was 78 months in non-shield-sites (NST-MM) patients and 20.5 months in shield-sites (ST-MM) patients ($p < 0.0001$); also in the long term, a better behavior was observed for NST-MM (80 vs. 42 months; $p < 0.0001$). Among 87 melanoma patients with a recent history of MM (≤ 30 days), we found that ST-MM patients showed lower values of vitamin D compared with NST-MM patients. Regarding BRAF status, a BRAF mutation was present in 13 % of ST-MM and in 41 % of NST-MM. Performing Mc-Nemar test, we found a statistical significant correlation between low serum levels of vitamin D in ST-MM and low percentage of BRAF mutation ($p = 0.03$), as well as between serum levels of vitamin D and high percentage of BRAF mutation in NST-MM ($p < 0.001$). All these aspects confirm that in ST-MM, other pathways play pivotal points, if compared with NST-MM.

Vulvar and Vaginal Melanomas-The Darker Shades of Gynecological Cancers.

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Melanomas of the skin are poorly circumscribed lesions, very frequently asymptomatic but unfortunately with a continuous growing incidence. In this landscape, one can distinguish melanomas originating in the mucous membranes and located in areas not exposed to the sun, namely the vulvo-vaginal melanomas. By contrast with cutaneous melanomas, the incidence of these types of melanomas is constant, being diagnosed in females in their late sixties. While hairy skin and glabrous skin melanomas of the vulva account for 5% of all cancers located in the vulva, melanomas of the vagina and urethra are particularly rare conditions. The location in areas less accessible to periodic inspection determines their diagnosis in advanced stages, often metastatic. Moreover, despite the large number of drugs newly approved in recent decades for the treatment of cutaneous melanoma, especially in the category of biological drugs, the mortality of vulvo-vaginal melanomas has remained almost constant. This, together with the absence of specific treatment guidelines due to the lack of a sufficient number of cases to conduct randomized clinical trials, makes melanomas with this localization a discouraging diagnosis, associated with a very poor prognosis. Our aim is therefore to draw attention to this oftentimes overlooked entity in order to encourage the community to employ various strategies meant to increase research in this area. By highlighting the main risk factors of vulvar and vaginal melanomas, as well as the clinical manifestations and molecular changes underlying these neoplasms, ideally novel therapeutic schemes will, in time, be brought into effect.